

SP ENGINEERING ACADEMY

Video Lectures / Course Materials

COMPLETE COURSE

DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

Part – I - Number System

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1. Number System

Video Link - <https://youtu.be/EXLTLeAggwk>

In digital electronics, the number system is used for representing the information. The number system has different base and the most common of them are the decimal, binary, octal, and hexadecimal. The base or radix of the number system is the total number of the ~~digit~~ digit used in the number system.

Suppose ~~the~~ if the number system representing the digit from 0 to 9, then the base of the system is the 10.

Types of Number Systems

(i) Decimal Number System

0, 1, 2, 3, 4, 5, 6, 7, 8, 9 - Base 10

(ii) Binary Number System

0, 1 - Base 2

Eg. $(0110)_2$, $(1001)_2$, $(1011)_2$

(iii) Octal Numbers

0, 1, 2, 3, 4, 5, 6, 7 - Base 8

All these digits from 0 to 7 have the same physical meaning as by decimal symbols, the next digit in the octal number is represented by 10, 11, 12, which are equivalent to decimal digits 8, 9, 10 respectively.

(iv) Hexadecimal Numbers

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F - Base 16

Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Decimal to Binary

$$\begin{array}{r} 1 \\ \hline 2 \overline{) 1} \\ \underline{0-1} \uparrow \end{array} \quad 01 \quad \underline{0001}$$

$$\begin{array}{r} 2 \\ \hline 2 \overline{) 2} \\ \underline{1-0} \uparrow \end{array} \quad 10 \quad \underline{0010}$$

$$\begin{array}{r} 3 \\ \hline 2 \overline{) 3} \\ \underline{1-1} \uparrow \end{array} \quad 11 \quad \underline{00101}$$

$$\begin{array}{r} 4 \\ \hline 2 \overline{) 4} \\ 2 \overline{) 2-0} \uparrow \\ \underline{1-0} \end{array} \quad 100 \quad \underline{0100}$$

$$\begin{array}{r} 5 \\ \hline 2 \overline{) 5} \\ 2 \overline{) 2-1} \uparrow \\ \underline{1-0} \end{array} \quad 101 \quad \underline{0101}$$

6

$$\begin{array}{r} 2 \overline{) 6} \\ 2 \overline{) 3-0} \\ \underline{1-1} \end{array} \uparrow$$

110

0110

7

$$\begin{array}{r} 2 \overline{) 7} \\ 2 \overline{) 3-1} \\ \underline{1-1} \end{array} \uparrow$$

111

0111

8

$$\begin{array}{r} 2 \overline{) 8} \\ 2 \overline{) 4-0} \\ 2 \overline{) 2-0} \\ \underline{1-0} \end{array} \uparrow$$

1000

1000

9

$$\begin{array}{r} 2 \overline{) 9} \\ 2 \overline{) 4-1} \\ 2 \overline{) 2-0} \\ \underline{1-0} \end{array} \uparrow$$

1001

1001

10

$$\begin{array}{r} 2 \overline{) 10} \\ 2 \overline{) 5-0} \\ 2 \overline{) 2-1} \\ \underline{1-0} \end{array} \uparrow$$

1010

1010

11

$$\begin{array}{r} 2 \overline{) 11} \\ 2 \overline{) 5-1} \\ 2 \overline{) 2-1} \\ \underline{1-0} \end{array} \uparrow$$

1011

1011

12

$$\begin{array}{r} 2 \overline{) 12} \\ 2 \overline{) 6-0} \\ 2 \overline{) 3-0} \\ \underline{1-1} \end{array} \uparrow$$

1100

1100

13

$$\begin{array}{r} 2 \overline{) 13} \\ 2 \overline{) 6-1} \\ 2 \overline{) 3-0} \\ \underline{1-1} \end{array} \uparrow$$

1101

1101

14

$$2 \overline{) 14}$$

$$2 \overline{) 7-0}$$

$$2 \overline{) 3-1}$$

$$1-1$$

1110

1110

15

$$2 \overline{) 15}$$

$$2 \overline{) 7-1}$$

$$2 \overline{) 3-1}$$

$$1-1$$

1111

1111

Decimal to Octal

0 → 0

1 → 1

2 → 2

3 → 3

4 → 4

5 → 5

6 → 6

7 → 7

8

$$8 \overline{) 8} \\ 1-0$$

8 → 10

9

$$8 \overline{) 9} \\ 1-1$$

9 → 11

10

$$8 \overline{) 10} \\ 1-2$$

10 → 12

11

$$8 \overline{) 11} \\ 1-3$$

11 → 13

12

$$8 \overline{) 12} \\ 1-4$$

12 → 14

$$13 \quad 8 \overline{) 13} \quad 1-5 \uparrow \quad \underline{13 \rightarrow 15}$$

$$14 \quad 8 \overline{) 14} \quad 4-6 \uparrow \quad \underline{14 \rightarrow 16}$$

$$15 \quad 8 \overline{) 15} \quad 1-7 \uparrow \quad \underline{15 \rightarrow 17}$$

2. Number Base Conversions

(i) Integral Part

Video Link - <https://youtu.be/3OBM9l1EepQ>

(a) Decimal to Binary

Eg 1 $(21)_{10} \rightarrow (10101)_2$

$$2 \overline{) 21}$$

$$2 \overline{) 10} - 1$$

$$2 \overline{) 5} - 0$$

$$2 \overline{) 2} - 1$$

$$1 - 0$$

$$10101$$

$$\downarrow \quad \downarrow$$

$$\text{MSB} \quad \text{LSB}$$

LSB - Least Significant bit

MSB - Most Significant bit

Eg 2 $(53.625)_{10} \rightarrow (110101.101)_2$

$$2 \overline{) 53}$$

$$2 \overline{) 26} - 1$$

$$2 \overline{) 13} - 0$$

$$2 \overline{) 6} - 1$$

$$2 \overline{) 3} - 0$$

$$1 - 1$$

$$0.625 \times 2 = 1.25 - 1$$

$$0.25 \times 2 = 0.5 - 0$$

$$0.5 \times 2 = 1 - 1$$

$$(110101.101)_2$$

Eg 3 $(112)_{10} \rightarrow (1110000)_2$

$$2 \overline{) 112}$$

$$2 \overline{) 56} - 0$$

$$2 \overline{) 28} - 0$$

$$2 \overline{) 14} - 0$$

$$2 \overline{) 7} - 0$$

$$2 \overline{) 3} - 1$$

$$1 - 1$$

$$(1110000)_2$$

Eg 4

$$(124.8125)_{10} \rightarrow (1111100.11010)_2$$

$$\begin{array}{r} 2 \overline{)124} \\ 2 \overline{)62} - 0 \\ 2 \overline{)31} - 0 \\ 2 \overline{)15} - 0 \\ 2 \overline{)7} - 1 \\ 2 \overline{)3} - 1 \\ 1 - 1 \end{array}$$

$$\begin{array}{l} 0.8125 \times 2 = 1.6250 \rightarrow 1 \\ 0.625 \times 2 = 1.250 \rightarrow 1 \\ 0.25 \times 2 = 0.5 \rightarrow 0 \\ 0.5 \times 2 = 1.0 \rightarrow 1 \\ 0 \times 2 = 0 \rightarrow 0 \checkmark \end{array}$$

$$(1111100.11010)_2$$

(b) Decimal to octal

Video Link - <https://youtu.be/f5EFi89w2HI>

Eg 1

$$(153)_{10} \rightarrow (231)_8$$

$$\begin{array}{r} 8 \overline{)153} \\ 8 \overline{)19} - 1 \\ 2 - 3 \end{array} \quad (231)_8$$

Eg 2

$$(926)_{10} \rightarrow (1636)_8$$

$$\begin{array}{r} 8 \overline{)926} \\ 8 \overline{)105} - 6 \\ 8 \overline{)14} - 3 \\ 1 - 6 \end{array} \quad (1636)_8$$

Eg 3

$$(645.926)_{10} \rightarrow (1205.7320)_8$$

$$\begin{array}{r} 8 \overline{)645} \\ 8 \overline{)80} - 5 \\ 8 \overline{)10} - 0 \\ 1 - 2 \end{array}$$

$$\begin{array}{l} 0.926 \times 8 = 7.408 \rightarrow 7 \\ 0.408 \times 8 = 3.264 \rightarrow 3 \\ 0.264 \times 8 = 2.112 \rightarrow 2 \\ 0.112 \times 8 = 0.896 \rightarrow 0 \checkmark \end{array}$$

(c) Decimal to Hexadecimal

Eg 1

$$(464.650)_{10} \rightarrow (1D0.A66)_{16}$$

$$16 \overline{) 10} \rightarrow A$$

$$16 \overline{) 464}$$

Video Link - <https://youtu.be/FNYNuA6IG3Q>

$$11 \rightarrow B$$

$$16 \overline{) 29} - 0$$

$$12 \rightarrow C$$

$$1 - 13(D)$$

$$13 \rightarrow D$$

$$14 \rightarrow E$$

$$0.650 \times 16 = 10.4 \rightarrow 10(A)$$

$$15 \rightarrow F$$

$$0.4 \times 16 = 6.4 \rightarrow 6$$

$$16$$

$$0.4 \times 16 = 6.4 \rightarrow 6$$

$$(1D0.A66)_{16}$$

Eg 2

$$(32064.1982)_{10} \rightarrow (7D40.32BD4)_{16}$$

$$16 \overline{) 32064}$$

$$0.1982 \times 16 = 3.1712 \rightarrow 3$$

$$16 \overline{) 2004} - 0$$

$$0.1712 \times 16 = 2.7392 \rightarrow 2$$

$$16 \overline{) 125} - 4$$

$$0.7392 \times 16 = 11.8272 \rightarrow 11(B)$$

$$7 - 13(D)$$

$$0.8272 \times 16 = 13.2352 \rightarrow 13(D)$$

$$0.2352 \times 16 = 3.7632 \rightarrow 3$$

$$7D40$$

$$32BD4$$

$$(7D40.32BD4)_{16}$$

Number Base Conversions

(ii) Successive Multiplication

(a) Binary to Decimal

Video Link - <https://youtu.be/c5OXZDU4z9E>

Eg 1

$$(101111)_2 \rightarrow (47)_{10}$$

$$1 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1$$

$$2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$$

$$1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$2^5 + 0 + 2^3 + 2^2 + 2^1 + 2^0$$

$$32 + 0 + 8 + 4 + 2 + 1$$

$$(47)_{10}$$

Eg 2

$$(101.1101)_2 \rightarrow (5.8125)_{10}$$

$$\begin{array}{ccccccc} 1 & 0 & 1 & . & 1 & 1 & 0 & 1 \\ 2^2 & 2^1 & 2^0 & & 2^{-1} & 2^{-2} & 2^{-3} & 2^{-4} \end{array}$$

$$1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times \frac{1}{2} + 1 \times \frac{1}{4} + 0 + 1 \times \frac{1}{16}$$

$$2^2 + 0 + 2^0 + \frac{1}{2} + \frac{1}{4} + \frac{1}{16}$$

$$4 + 1 + \frac{8 + 4 + 1}{16}$$

$$5 + \frac{13}{16}$$

$$(5.8125)_{10}$$

$$\begin{array}{r} 0.8125 \\ 16 \overline{) 130} \\ \underline{128} \end{array}$$

$$\begin{array}{r} 20 \\ 16 \\ \underline{40} \\ 32 \\ \underline{80} \\ 30 \end{array}$$

(b) Octal to Decimal

Video Link - <https://youtu.be/CIh8o93Pdel>

Eg 1

$$(237.45)_8 \xrightarrow{159} (159.578)_{10}$$

$$\begin{array}{ccccccc} 2 & 3 & 7 & . & 4 & 5 \\ 8^2 & 8^1 & 8^0 & & 8^{-1} & 8^{-2} \end{array}$$

$$2 \times 8^2 + 3 \times 8^1 + 7 \times 8^0 + 4 \times 8^{-1} + 5 \times 8^{-2}$$

$$\begin{array}{c} 64 \\ 2 \times 16 \end{array} + 3 \times 8 + 7 \times 1 + \frac{4}{8} + \frac{5}{64}$$

$$\begin{array}{c} 128 \\ 32 \end{array} + 24 + 7 + 0.5 + 0.078$$

$$159 + 0.578$$

$$\begin{array}{c} 159 \\ (159.578)_{10} \end{array}$$

Eg 2 $(121.231)_8 \rightarrow (81.298775)_{10}$

1	2	1	.	2	3	1
8^2	8^1	8^0	.	8^{-1}	8^{-2}	8^{-3}

$$1 \times 8^2 + 2 \times 8^1 + 1 \times 8^0 + 2 \times 8^{-1} + 3 \times 8^{-2} + 1 \times 8^{-3}$$

$$\begin{array}{l} 64 \\ 1 \times \cancel{64} + 2 \times 8 + 1 \times 8^0 + \frac{2}{8} + \frac{3}{\cancel{64}} + \frac{1}{512} \end{array}$$

$$\begin{array}{l} 64 \\ \cancel{64} + 16 + \bullet \end{array} \cdot \begin{array}{l} 0.25 \\ \cancel{0.5} \end{array} + \begin{array}{l} 0.046875 \\ \cancel{0.1875} \end{array} + 0.0019$$

$$(81.298775)_{10}$$

Video Link - <https://youtu.be/3kk2uMIhPmM>

(c) Hexadecimal to Decimal

Eg 1 $(2F3)_{16} \rightarrow (755)_{10}$

2	F	3
2	15	3
16^2	16^1	16^0

$$2 \times 16^2 + 15 \times 16^1 + 3 \times 16^0$$

$$2 \times 256 + 15 \times 16 + 3 \times 1$$

$$512 + 240 + 3$$

$$(755)_{10}$$

Eg 2 $(237.45)_{16} \rightarrow (567.26)_{10}$

2	3	7	.	4	5
16^2	16^1	16^0	.	16^{-1}	16^{-2}

$$2 \times 16^2 + 3 \times 16^1 + 7 \times 16^0 + 4 \times 16^{-1} + 5 \times 16^{-2}$$

$$2 \times 256 + 3 \times 16 + 7 \times 1 + \frac{4}{16} + \frac{5}{256}$$

$$512 + 48 + 7 + 0.25 + 0.01$$

$$(567.26)_{10}$$

(iii) Grouping Techniques - Number Base Conversions

(a) Binary to Octal

Eg 1 $(111\ 111\ 110)_2 \rightarrow (376)_8$

011	111	110	8421
421	421	421	421
	3	7	6
			$(376)_8$

Video Link <https://youtu.be/bYvPTYQCNlg>

Eg 2 $(1110111111)_2 \rightarrow (1677)_8$

001	110	111	111
421	421	421	421
1	6	7	7
			$(1677)_8$

Video Link - https://youtu.be/enxGj9Sej_E

(b) Binary to Hexadecimal

Eg 1 $(11011111\ 0110)_2 \rightarrow (DF6)_{16}$

1101	1111	0110
8421	8421	8421
13	15	6
D	F	6
		$(DF6)_{16}$

Eg 2 $(10111001\ 01)_2 \rightarrow (2E5)_{16}$

0010	1110	0101
8421	8421	8421
2	13	5
2	E	5
		$(2E5)_{16}$

(c) Octal to Binary

Eg 1 $(376)_8 \rightarrow (01111110)_2$

$$\begin{array}{ccc} 3 & 7 & 6 \\ 011 & 111 & 110 \end{array}$$

$$\begin{array}{ccc} 2 \overline{)3} & 2 \overline{)7} & 2 \overline{)6} \\ \underline{1-1} & 2 \overline{)3-1} & 2 \overline{)3-0} \\ & 1-1 & 1-0 \end{array}$$

$$(01111110)_2$$

Eg 2 $(7642)_8 \rightarrow (111110100010)_2$

$$\begin{array}{cccc} 7 & 6 & 4 & 2 \\ 111 & 110 & 100 & 010 \end{array}$$

$$\begin{array}{cccc} 2 \overline{)7} & 2 \overline{)6} & 2 \overline{)4} & 2 \overline{)2} \\ 2 \overline{)3-1} & 2 \overline{)3-0} & 2 \overline{)2-0} & \underline{1-0} \\ 1-1 & 1-1 & 1-0 & \end{array}$$

$$(111110100010)_2$$

Video Link - <https://youtu.be/myUXOAe43fc>(d) Hexadecimal to Binary

Eg 1 $(2D5)_{16} \rightarrow (001011010101)_2$

$$\begin{array}{ccc} 2 & D & 5 \\ 2 & 13 & 5 \\ 0010 & 1101 & 0101 \end{array}$$

$$\begin{array}{ccc} 2 \overline{)2} & 2 \overline{)13} & 2 \overline{)5} \\ 1-0 & 2 \overline{)6-1} & 2 \overline{)2-1} \\ & 2 \overline{)3-0} & 1-0 \\ & 1-1 & \end{array}$$

$$(001011010101)_2$$

Eg 2 $(BE2A)_{16} \rightarrow (1011111000101010)_2$

$$\begin{array}{cccc} B & E & 2 & A \\ 11 & 14 & 2 & 10 \\ 1011 & 1110 & 0010 & 1010 \end{array}$$

$$\begin{array}{cccc} 2 \overline{)11} & 2 \overline{)14} & 2 \overline{)2} & 2 \overline{)10} \\ 2 \overline{)5-1} & 2 \overline{)7-0} & 1-0 & 2 \overline{)5-0} \\ 2 \overline{)2-1} & 2 \overline{)3-1} & & 2 \overline{)2-1} \\ 1-0 & 1-1 & & 1-0 \end{array}$$

$$(1011111000101010)_2$$

(e) Hexadecimal to Octal

Eg 1 $(47)_{16} \rightarrow (01000111)_2 \rightarrow (107)_8$

Hexa to Binary

$$\begin{array}{cc} 4 & 7 \\ 0100 & 0111 \\ (01000111)_2 \end{array}$$

$$\begin{array}{cc} 2 \overline{)4} & 2 \overline{)7} \\ 2 \overline{)2-0} & 2 \overline{)3-1} \\ 1-0 & 1-1 \end{array}$$

Binary to Octal

$$\begin{array}{ccc} 001 & 000 & 111 \\ 421 & 421 & 421 \\ 1 & 0 & 7 \\ (107)_8 \end{array}$$

Eg 2 $(A7B9)_{16} \rightarrow (1010011110111001)_2 \rightarrow (123671)_8$

Hexa to Binary

$$\begin{array}{cccc} A & 7 & B & 9 \\ 10 & 7 & 11 & 9 \\ 1010 & 0111 & 1011 & 1001 \end{array}$$

$$\begin{array}{cccc} 2 \overline{)10} & 2 \overline{)7} & 2 \overline{)11} & 2 \overline{)9} \\ 2 \overline{)5-0} & 2 \overline{)3-1} & 2 \overline{)5-1} & 2 \overline{)4-1} \\ 2 \overline{)2-1} & 1-1 & 2 \overline{)2-1} & 2 \overline{)2-0} \\ 1-0 & & 1-0 & 1-0 \end{array}$$

$$(1010011110111001)_2$$

Binary to Octal

$$\begin{array}{cccccc} 001 & 010 & 011 & 110 & 111 & 001 \\ 421 & 421 & 421 & 421 & 421 & 421 \\ 1 & 2 & 3 & 6 & 7 & 1 \\ (123671)_8 \end{array}$$

(f) Octal to Hexadecimal

Eg 1 $(32)_8 \rightarrow (011010)_2 \rightarrow (1A)_{16}$

Octal to Binary

Video Link - https://youtu.be/_S4NC1Qhhkc

$$\begin{array}{cc} 3 & 2 \\ 011 & 010 \end{array}$$

$$(011010)_2$$

$$\begin{array}{r} 2 \overline{) 3} \quad 2 \overline{) 2} \\ 1-1 \quad 1-0 \end{array}$$

Binary to Hexadecimal

$$\begin{array}{cc} 0001 & 1010 \\ 8421 & 8421 \end{array}$$

$$\begin{array}{cc} 1 & 10 \end{array}$$

$$\begin{array}{cc} 1 & A \end{array}$$

$$(1A)_{16}$$

Eg 2 $(746)_8 \rightarrow (111100110)_2 \rightarrow (1E6)_{16}$

Octal to Binary

$$\begin{array}{ccc} 7 & 4 & 6 \\ 111 & 100 & 110 \end{array}$$

$$(111100110)_2$$

$$\begin{array}{r} 2 \overline{) 7} \quad 2 \overline{) 4} \quad 2 \overline{) 6} \\ 2 \overline{) 3-1} \quad 2 \overline{) 2-0} \quad 2 \overline{) 3-0} \\ 1-1 \quad 1-0 \quad 1-1 \end{array}$$

Binary to Hexadecimal

~~11110011~~

$$\begin{array}{ccc} 0001 & 1110 & 0110 \\ 8421 & 8421 & 8421 \end{array}$$

$$\begin{array}{ccc} 1 & 14 & 6 \end{array}$$

$$\begin{array}{ccc} 1 & E & 6 \end{array}$$

$$(1E6)_{16}$$